CS-008 Resins, Mixing & Cure Schedules

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Title	Resins, Mixing & Cure Schedules
Revision	A
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Supersedes	Ratio/pot-life tribal knowledge

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted w/ Claude — SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline — the one-page ratio/cure table every other standard cites

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1. Purpose

One table for every resin we stock: ratios, pot lives, cure/demould times — plus the mixing discipline and exotherm rules that apply to all of them. Exists because of soft-spot failures (bad ratios), the first forged-CF part cracking from resin starvation, and a few too many hot cups.

2. Scope

All team resin systems. Process-specific use lives in CS-004 (XCR), CS-006 (IN2), CS-007 (West 105).

3. References

Ref	Document	Where
R1	IN2 TDS · R2 XCR TDS · R3 West 105/205, /206, /209 TDS	`04 Datasheets/` (see INDEX.md)
R4	SDS for all of the above	`04 Datasheets/`
R5	*Forged/Compression Molded CF* study (mix math, starvation failure)	`SN5 Composites/Documentation/Forged_Compressio` `Molded Carbon Fiber.docx`

4. Definitions

Pot life (usable time in the pot at stated mass/temp), gel, demould, full cure, exotherm.

5. Equipment & Materials — THE TABLE (verify each against its TDS before signing)

System	Ratio (by weight)	Pot life	Demould	Full properties	Use
IN2 + AT30 SLOW	100:30	120–150 min @25 °C	18–24 h	14 d @25 °C	Infusion default (CS-006)
IN2 + AT30 FAST	100:30	10–14 min @25 °C	6–8 h	14 d	Small infusions only
West 105 + 206 SLOW	5.36:1 (5:1 by volume pumps)	20–25 min @72 °F	thin-film solid 10–15 h	days	Wet layup default (CS-007)
West 105 + 205 FAST	5.19:1 (5:1 by volume)	9–12 min @72 °F	6–8 h	days	Small wet layups, tabs
West 105 + 209	**3.68:1 (3:1 by volume — DIFFERENT pump ratio than 205/206!)**	40–50 min @72 °F (100 g)	thin-film solid 20–24 h	max strength 4–9 d; min temp 70 °F	Hot-day / very large wet layups
XCR	100:41 (2:1 by volume)	8 min @20 °C (150 g)	8.5 h initial	14 d @20 °C	Mold sealing (CS-004)

6. Safety — exotherm and sensitization (applies to every system)

- All epoxies here are **skin/respiratory sensitizers** (R4): nitrile gloves, ventilation, wash skin with soap not solvent, A-filter respirator for long/enclosed sessions.
- **Exotherm rules:** mix the smallest batch that does the job; pour out of the pot promptly (mass concentrated in a cup accelerates); a warm cup = use immediately or spread thin; a hot/smoking cup = carry outside onto concrete and leave it — do not bin, do not water-dilute indoors. Nothing goes in the trash until fully cured and cool.
- Never mix off-ratio to "speed up" or "slow down" cure — blend hardener *speeds* (same total ratio, per R1/R3) instead; off-ratio = uncured sticky part.

7. Procedure (outline)

1. Pick the system from §5 by process standard; confirm pot life fits the job + a margin. 209 uses **3:1 pumps**, not the 5:1 pumps used by 205/206 — check the pump head before stroking, this is the classic wrong-ratio trap.
2. Weigh (scale, 1 g) or pump (West, full strokes only); record masses on the WO.
3. Mix 2 min scraping walls/base; **double-pot** (transfer + remix, fresh stick) for anything going on a part surface.
4. Forged CF mix math (R5): carbon mass mold volume × 1.4 g/cc; resin fiber × 1.25 — starving the mix made the first part brittle (failed bend test); don't eyeball it.
5. Cure at 20 °C wherever possible; log ambient temp on the WO when it isn't (cold cures run long — see each TDS).
6. Leftovers: spread thin on scrap or leave outside to cure; label per CS-001.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Ratio	Weighed/pumped per §5, recorded	Scale/pump count	WO
No soft spots at demould	Fingernail test on flange/trim zone	Manual	WO qualityChecks

9. Records

Batch masses, system + hardener speed, ambient temp, cure duration — on every WO that consumes resin.